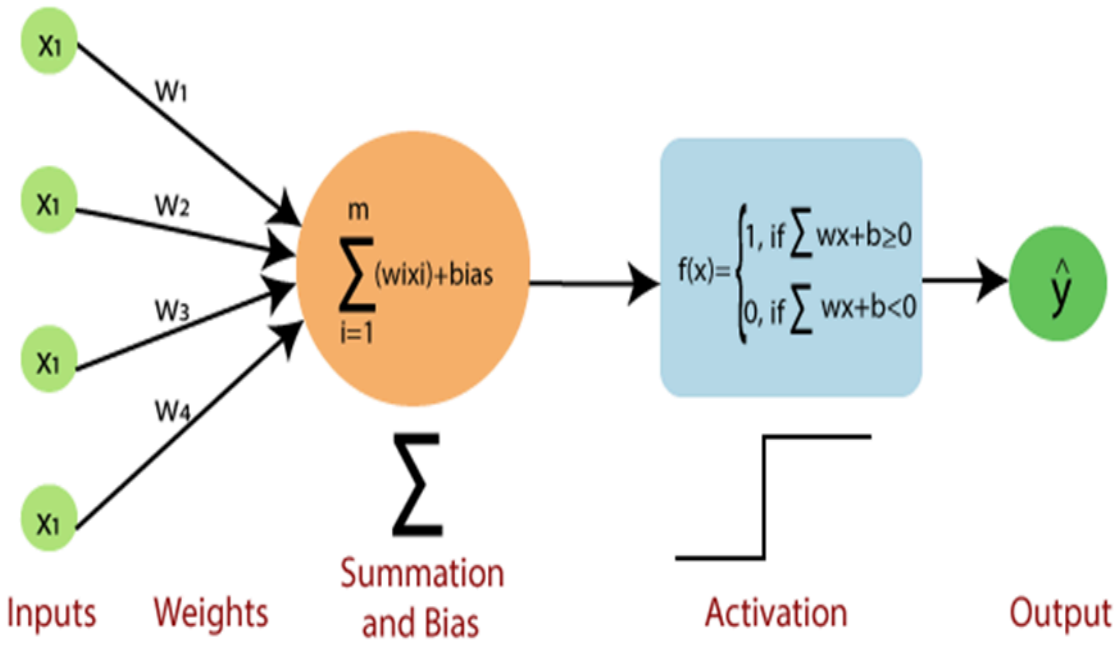
Perceptrons & Activation Functions: Foundations of Neural Networks

Overview

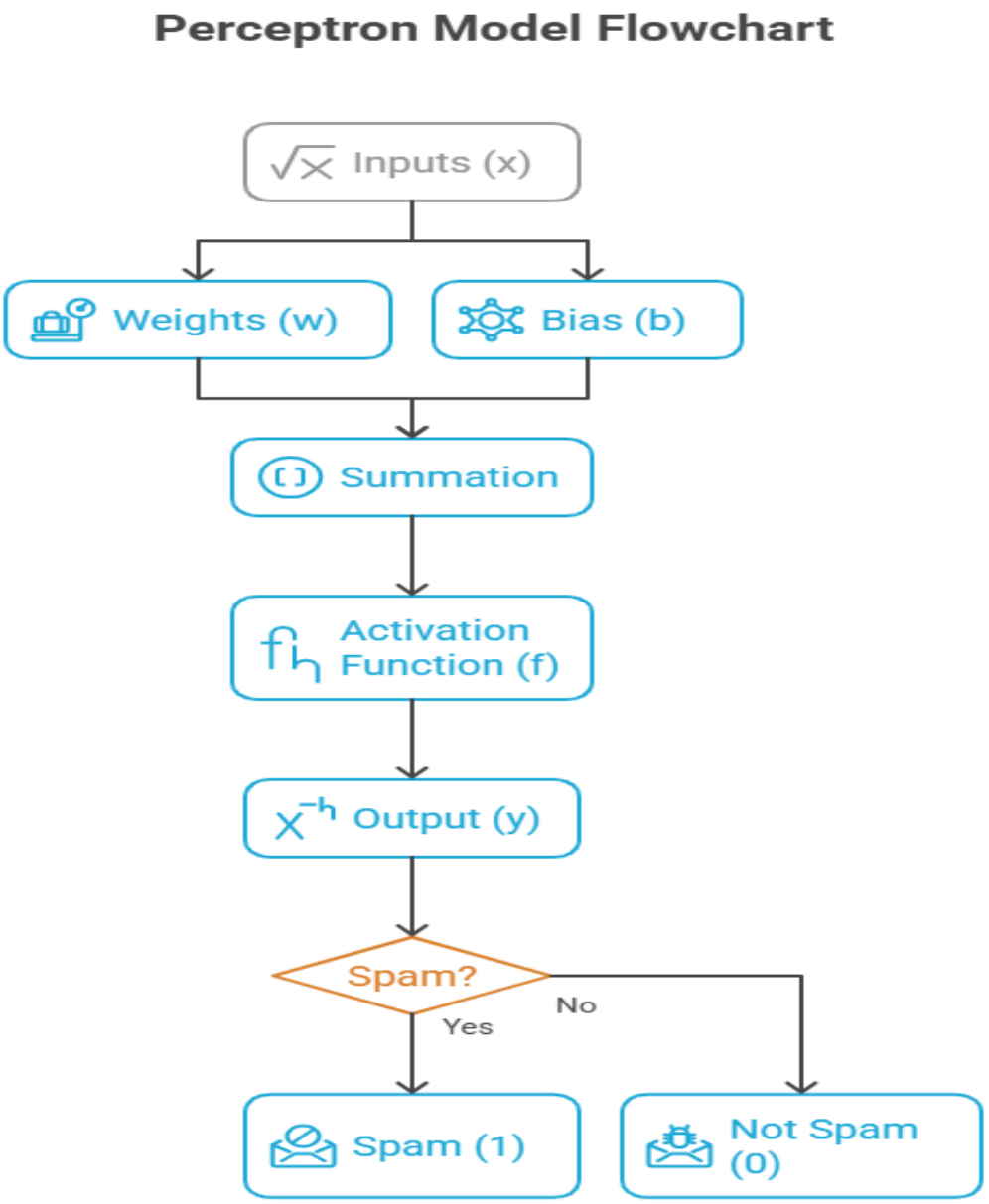
The **Perceptron** is the building block of a neural network. It mimics a biological neuron:

- Inputs come in (like signals to dendrites).
- Weights control the importance of each input.
- A bias shifts the decision boundary.
- An **activation function** decides the output.



Without activation functions, neural networks would just be **linear models** — they wouldn't capture complex patterns.

Perceptron Model



Equation:

y=f(∑wixi+b)y = f\Big(\sum w_i x_i + b\Big)y=f(∑wixi+b)

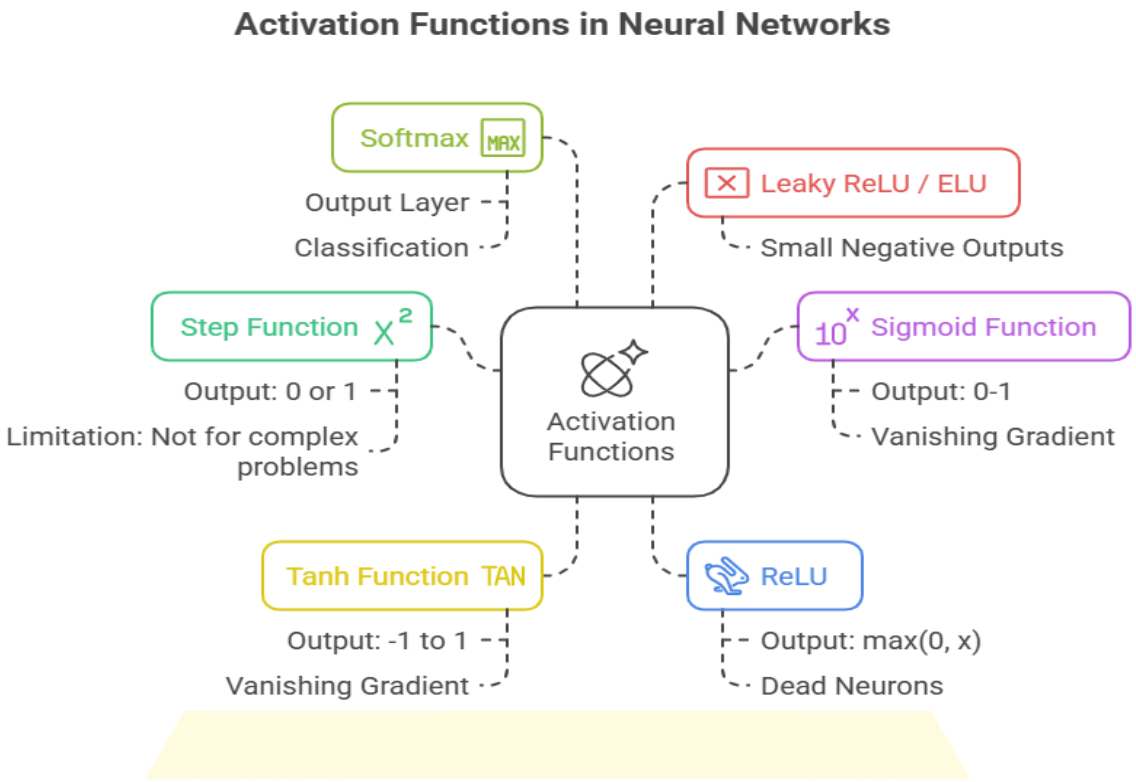
- **Inputs (x)**: Features of data (e.g., pixels of an image).
- **Weights (w)**: Learnable parameters (importance of each input).

- **Bias (b)**: Adjusts decision threshold.

- **Activation (f)**: Non-linear function that decides the output.
  Example: Spam email filter
- Inputs = features like "word frequency," "contains links," etc.

- Perceptron learns weights to decide: Spam (1) or Not Spam (0).

Activation Functions



1. **Step Function**

   - Output is 0 or 1.

   - Used in the first perceptrons.

   - Limitation: Not useful for complex problems.

2. **Sigmoid Function**

   f(x)=11+e−xf(x) = \frac{1}{1+e^{-x}}f(x)=1+e−x1

   - Smooth, outputs between 0–1.
   - Good for probabilities.
   - Vanishing gradient problem.

3. **Tanh Function**

   - Output between -1 and 1.
   - Centered around 0 (better than sigmoid).
   - Still suffers from vanishing gradients.

4. **ReLU (Rectified Linear Unit)**

   f(x)=max(0,x)f(x) = \max(0, x)f(x)=max(0,x)

   - Fast and simple.
   - Most widely used in DL.
   - Dead neurons (if too many outputs = 0).

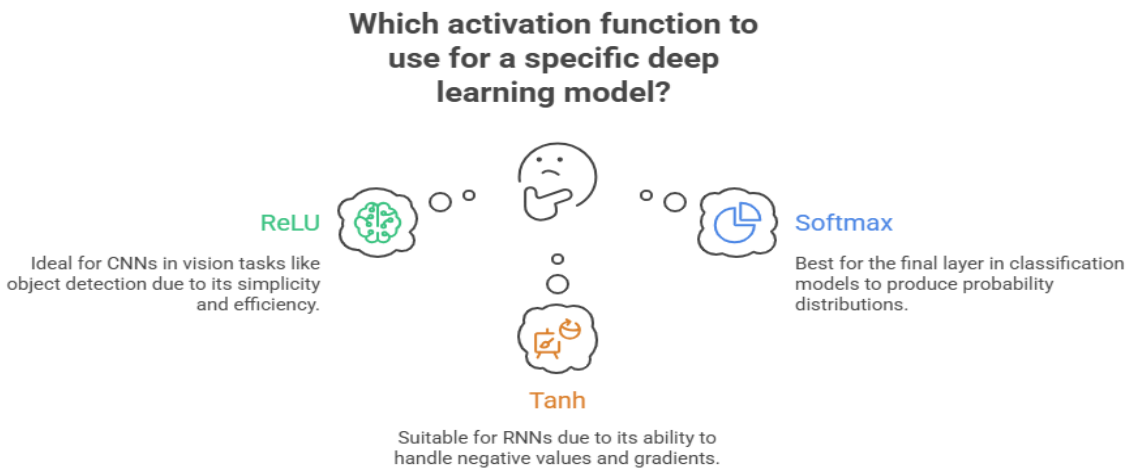
5. **Leaky ReLU / ELU**

   - Fix dead neuron issues by allowing small negative outputs.

6. **Softmax**

   - Converts scores into probability distribution.
   - Used in the **output layer** for classification.

DL Applications



- ReLU → CNNs for vision (e.g., object detection).
- Softmax → Final layer in classification models.
- Tanh → Used in RNNs.